Multi-Parameter Persistent Homology Reveals Income-Stratified Topological Structure in UK Employment Trajectories

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Abstract

Employment trajectory topology, as revealed by persistent homology of Vietoris–Rips filtrations, captures structural features of career pathways. Single-parameter persistent homology, however, treats all trajectories identically regardless of income. We introduce a bifiltration on British Household Panel Survey and Understanding Society data (1991–2023, $N = 27{,}280$), where simplices are filtered simultaneously by pairwise distance in PCA-embedded trajectory space and by an income threshold parameter. The signed measure decomposition from multi-parameter persistent homology reveals that H_1 topological features — loops representing cyclical or multi-pathway career structures — concentrate at mid-to-high income levels. Quartiles Q3 and Q4 carry approximately four times the H_1 mass of Q1, and this stratification is significant under 999 income-label permutations ($p < 0.001$). Single-parameter persistent homology on the same landmarks shows no income gradient in H_1 feature density, confirming that the bifiltration captures structure invisible to standard topological methods. Complementary invariants — the Hilbert function and the rank invariant — confirm the income asymmetry

and reveal that all H_1 features persist to infinity along the income axis, ruling out income-confined topological structures. We argue that the topological simplicity of low-income trajectory space — fewer loops, fewer alternative pathways — formalises a structural poverty trap: limited career diversity constrains escape routes from low-income states.

Keywords: multi-parameter persistent homology, signed measures, bifiltration, poverty traps, topological data analysis, employment trajectories, BHPS, Understanding Society

1 Introduction

1.1 Motivation: The Limitations of Single-Parameter Topology

Paper 1 in this programme applied Vietoris–Rips persistent homology to UK employment trajectories embedded in a 20-dimensional PCA space, finding non-trivial H_0 and H_1 structure that exceeds the expectations of first-order Markov null models. That analysis uses a single filtration parameter — the Rips radius — which treats all pairwise distances equally regardless of economic outcome. A trajectory cluster in a low-income region of the space has the same topological weight as one in a high-income region.

Poverty trap theory (Azariadis and Stachurski, 2005; Barrett and Carter, 2013) predicts structurally distinct configurations at low income levels — absorbing states, threshold effects, bifurcation points. Single-parameter PH cannot test this prediction because it lacks an income axis. The question motivating this paper is whether the *topological* structure of trajectory space varies systematically with income.

1.2 Multi-Parameter Persistence: From Theory to Computation

Multi-parameter persistent homology extends single-parameter PH by indexing the filtration by two or more real-valued parameters simultaneously. The mathemati-

cal theory was established by Carlsson and Zomorodian (2009), who proved that no complete barcode decomposition exists in the multi-parameter case — unlike single-parameter PH, where the barcode is a complete invariant. This negative result stalled computational applications for over a decade.

Recent advances have made multi-parameter PH practical. The signed barcode decomposition (Botnan et al., 2022) and its computational implementation in the `multipers` library (Loiseaux et al., 2023, 2024) provide tractable invariants — signed measures, Hilbert functions, rank invariants — that capture the essential multi-parameter structure. This paper is among the first applications of these tools to social science panel data.

1.3 Research Questions

(1) Does the Rips $\times$ income bifiltration reveal topological structure absent from single-parameter Rips PH?

(2) How does H_1 topological complexity distribute across income levels, and is this distribution significantly different from what distance structure alone would predict?

(3) What does income-stratified topology tell us about the structure of poverty traps?

(4) Do H_1 features born at low income persist across the full income range, or are they confined to low-income regions of the bifiltration?

1.4 Paper Roadmap

Section 2 reviews the poverty trap and multi-parameter PH literatures. Section 3 describes the bifiltration construction, signed measure computation, and permutation null testing. Section 4 presents results from the 999-permutation analysis, P01 baseline comparison, GMM regime overlay, and complementary invariants (Hilbert function and rank invariant). Section 5 discusses the interpretation of topological simplicity as a poverty trap signature.

2 Literature Review

2.1 Poverty Traps and Threshold Effects

Poverty trap theory posits the existence of multiple equilibria in income dynamics, where self-reinforcing mechanisms hold individuals below critical thresholds (Azariadis and Stachurski, 2005). Barrett and Carter (2013) distinguish structural traps — where the asset or human capital threshold is the binding constraint — from stochastic traps, where shocks push individuals below a threshold from which recovery is difficult. In the UK context, Shildrick et al. (2012) document persistent cycling between low-pay and no-pay states that resembles a structural trap without requiring a sharp threshold.

Markov models of income dynamics (Cappellari and Jenkins, 2004) capture transition probabilities but not the geometric structure of the trajectory space. The contribution of topological methods is to characterise the *shape* of career pathway space at each income level — not just transition rates, but the connectivity and cyclical structure of available pathways.

2.2 Single-Parameter Persistent Homology for Social Science

Paper 1 in this programme applied VR PH to UK employment trajectories, finding significant H_0 (cluster) and H_1 (loop) structure. The Markov memory ladder null test confirmed that trajectory topology exceeds first-order Markov expectations. However, single-parameter PH is outcome-agnostic: the filtration parameter is pairwise distance in the embedding, which carries no information about income.

Post-hoc stratification — computing PH separately on low-income and high-income subsets — changes the point cloud and thus the simplicial complex; it does not decompose the *joint* topology of the full space. Multi-parameter PH provides a principled framework for this decomposition.

2.3 Multi-Parameter Persistent Homology

The theoretical foundations of multi-parameter PH were established by Carlsson and Zomorodian (2009), who showed that while persistence modules over a totally ordered set decompose into interval summands (yielding the barcode), persistence modules over a partially ordered set — as in multi-parameter PH — do not admit such a decomposition in general.

Practical invariants for multi-parameter PH include: the rank invariant (Carlsson and Zomorodian, 2009), which is the complete discrete invariant for 2-parameter modules; the Hilbert function, which gives graded Betti numbers across the bifiltration grid; and the signed barcode/signed measure (Botnan et al., 2022; Loiseaux et al., 2023), which decomposes the persistence module into signed interval summands.

The `multipers` library (Loiseaux et al., 2024) implements these invariants with a scikit-learn-style API and `gudhi` backend. We use signed measures as the primary invariant and compute Hilbert functions and one-dimensional slices of the rank invariant as complementary diagnostics (Appendix B).

2.4 Why Signed Measures?

The signed measure μ_k for homology degree k is a discrete signed measure on the bifiltration parameter space that encodes where topological features are born (positive weight) and die (negative weight). For applied analysis, it has two advantages over the Hilbert function and rank invariant: (1) it is computationally the least expensive of the three, and (2) it admits a natural income-stratification decomposition by summing absolute weights within income bands.

3 Data and Methods

3.1 Data: BHPS/USoc Panel Trajectories

Source: British Household Panel Survey (1991–2008) and Understanding Society (2009–2023). **Sample:** $N = 27{,}280$ individuals with ≥ 5 waves of employment state observations, from the Paper 1 pipeline. **Embedding:** 4-gram frequency embedding followed by PCA to 20 dimensions; each individual is a point in $\mathbb{R}^{20}$.

Income variable: Mean income score derived from employment state sequences. Each state's final character encodes income band (L, M, H), mapped to a continuous score: $L = 0, M = 1, H = 2$, averaged across observed waves. The resulting per-person scalar $y_i \in [0, 2]$ serves as the second filtration parameter.

Note: This proxy income score tracks the trajectory-level income component rather than a survey-derived monetary income variable. For the methodological contribution of this paper — demonstrating that bifiltration PH reveals income-stratified topology — the proxy is adequate. A future analysis (Paper 5) will use equivalised net household income from the USoc derived variable `fihhmnnet1_dv`.

3.2 Bifiltration Construction

We construct a two-parameter filtration $\{K_{\varepsilon,\tau}\}$ where:

Parameter 1 — Rips distance (ε): Standard Vietoris–Rips filtration on the PCA embedding. An edge between individuals i and j appears when $\|x_i - x_j\| \leq \varepsilon$.

Parameter 2 — Income threshold (τ): A vertex i enters the complex when its mean income $y_i \leq \tau$. This builds the complex upward from the poorest trajectories.

Formally:

$$K_{\varepsilon,\tau} = \mathrm{VR}_\varepsilon\big(\{x_i \mid y_i \leq \tau\}\big). \tag{1}$$

Landmarking: We subsample 2,000 landmarks via greedy maxmin subsampling (seed = 42) from the 27,280-point embedding, preserving geometric coverage. The Rips radius is set at the 10th percentile of pairwise distances in a 500-point subsample

($\varepsilon_{\max} = 11.126$), yielding 215,958 simplices.

Income discretisation: The income axis τ is discretised using a **quantile grid**: 20 thresholds at uniform quantiles of the 2,000-landmark income distribution. Exploratory comparison (Appendix A) confirmed that quantile grids produce more stable H_1 low-income fraction estimates than fixed (uniform-spacing) grids: convergence to $\sim$0.33 by 10 grid steps (range $[0.33, 0.44]$) versus $[0.26, 0.51]$ for fixed grids. The Rips axis is discretised into 20 uniform steps from 0 to $\varepsilon_{\max}$.

3.3 Multi-Parameter Persistence Computation

The bifiltration is constructed via `multipers.filtrations.RipsLowerstar`, which builds a 2-parameter simplex tree from the point cloud and income function. The **signed measure** μ_k for homology degree $k \in \{0, 1\}$ is computed on the 20×20 quantile grid using `multipers.signed_measure()`.

We report three summary statistics:

1. **Income-quartile mass decomposition:** The τ axis is partitioned into four quartiles (using income percentiles of the full landmark set). Total absolute signed measure mass $\sum_{(e,t)\in Q_j} |\mu_k(e,t)|$ is computed for each quartile $j \in \{1, 2, 3, 4\}$.

2. H_1 **low-income fraction:** $f_{\text{low}} = \sum_{\tau \leq \text{med}} |\mu_1| \,/\, \sum |\mu_1|$.

3. **Q1/Q3 mass ratio:** $r = m_{Q1}/m_{Q3}$, comparing topological complexity in the poorest and second-richest quartiles. Under income-exchangeability, this should be close to 1.0.

3.4 Complementary Invariants

We compute two additional invariants using `multipers.signed_measure()` with the `invariant` parameter:

Hilbert function (`invariant="hilbert"`): The Hilbert function $\mathcal{H}_k(\varepsilon, \tau) = \dim H_k(K_{\varepsilon,\tau})$ counts features *alive* at each grid point, providing a cumulative "state" view comple-

mentary to the signed measure's "differential" (birth/death) view. We compute $\mathcal{H}_0$ and $\mathcal{H}_1$ on both 10×10 and 20×20 grids.

Rank invariant (`invariant="rank"`): The rank invariant captures which features persist across parameter ranges, returning 4-dimensional support points $(b_\varepsilon, b_\tau, d_\varepsilon, d_\tau)$ recording the birth and death coordinates of each feature in the bifiltration. We compute the H_1 rank invariant to test whether features born at low income survive to high income or are confined to low-income regions.

3.5 Permutation Null Testing

Null hypothesis: The income-stratified distribution of H_1 signed measure mass is fully explained by the distance structure — shuffling income labels does not change the income-quartile decomposition.

Procedure: For $b = 1, \ldots, 999$:

1. Shuffle income labels across the 2,000 landmarks (breaking the income-trajectory association while preserving the distance structure).
2. Reconstruct the bifiltration with shuffled incomes.
3. Compute the H_1 signed measure on the *same* quantile grid (derived from the original, unshuffled incomes).
4. Record $f_{\text{low}}^{(b)}$ and $r^{(b)}$.

***p*-value:** Two-sided:

$$\hat{p} = B^{-1} \sum_{b=1}^{B} \mathbb{1}\left[\left|T^{(b)} - \bar{T}\right| \geq \left|T_{\text{obs}} - \bar{T}\right|\right], \tag{2}$$

where $\bar{T}$ is the null mean and T is each test statistic.

The permutation preserves the distance matrix (same points in $\mathbb{R}^{20}$) and the income marginal distribution (same set of income values), but destroys the association between trajectory position and income level. A significant result means that the

observed income-topology association is not an artefact of the embedding geometry alone.

3.6 Comparison to Paper 1 Single-Parameter Baseline

We compute standard VR PH (ignoring income) on three subsets of the same 2,000 landmarks: the full set, the low-income subset ($y_i \leq Q_1$), and the high-income subset ($y_i \geq Q_3$). If single-parameter PH shows similar H_1 density across income subsets, this confirms that the income stratification is only visible through the bifiltration.

4 Results

4.1 Bifiltration Structure

The bifiltration on 2,000 maxmin landmarks from the 27,280 $\times$ 20D Paper 1 embedding yielded 215,958 simplices at Rips radius 11.126. Landmark income scores had mean 1.02, median 1.00, and standard deviation 0.48, centred around the midpoint of the $[0, 2]$ range.

The H_0 signed measure (229 support points on the quantile grid) showed roughly balanced income distribution (low-income fraction = 0.512). Connected components form at all income levels — the large-scale cluster structure of trajectory space is income-symmetric.

The H_1 signed measure (218 support points) showed pronounced income asymmetry: low-income fraction = 0.329, far below the 0.50 expected under income-exchangeability and below the H_0 baseline of 0.51.

4.2 Income-Quartile Decomposition of H_1 Topology

The H_1 signed measure mass decomposes across income quartiles as shown in Table 1.

Quartiles Q3 and Q4 carry 73.7% of total H_1 mass. The Q1/Q3 ratio is 0.258 — topological cycles are approximately four times more prevalent at Q3 income levels

Table 1: H_1 signed measure mass by income quartile (20×20 quantile grid, 2,000 landmarks).

Quartile	Range (income score)	$\lvert H_1$ mass$\rvert$	Share
Q1 (poorest)	0.00–0.60	20,336	9.6%
Q2	0.60–1.00	35,448	16.7%
Q3	1.00–1.36	78,674	37.1%
Q4 (richest)	1.36–2.00	77,524	36.6%

than at Q1.

Interpretation: Mid-to-high income trajectory space has rich topological structure: multiple career pathways (e.g., different combinations of employment, self-employment, reduced hours) coexist and form loops in the embedding. Low-income trajectory space is topologically simpler — fewer distinct pathways create less cyclical connectivity. This topological simplicity reflects the constrained option set that characterises structurally trapped trajectories.

4.3 Permutation Null Test Results

The 999 income-label permutations took 24 minutes on an 8-core i7 (parallelised via `joblib`). Results are shown in Table 2.

Table 2: Permutation null test results (999 income-label shuffles).

Statistic	Observed	Null mean	p-value
H_1 low-income fraction	0.329	0.265	< 0.001
Q1/Q3 mass ratio	0.258	0.161	< 0.001

Both test statistics are significantly different from the permutation null ($p < 0.001$). The direction of the effect is notable: the observed low-income fraction (0.329) is *higher* than the null mean (0.265), and the observed Q1/Q3 ratio (0.258) is higher than the null mean (0.161).

Interpretation: Under random income assignment, H_1 mass concentrates even more extremely at high τ than it does with real income labels. The real income-trajectory association *moderates* the asymmetry — low-income trajectories contribute more H_1 complexity than random assignment would predict. This is consistent with

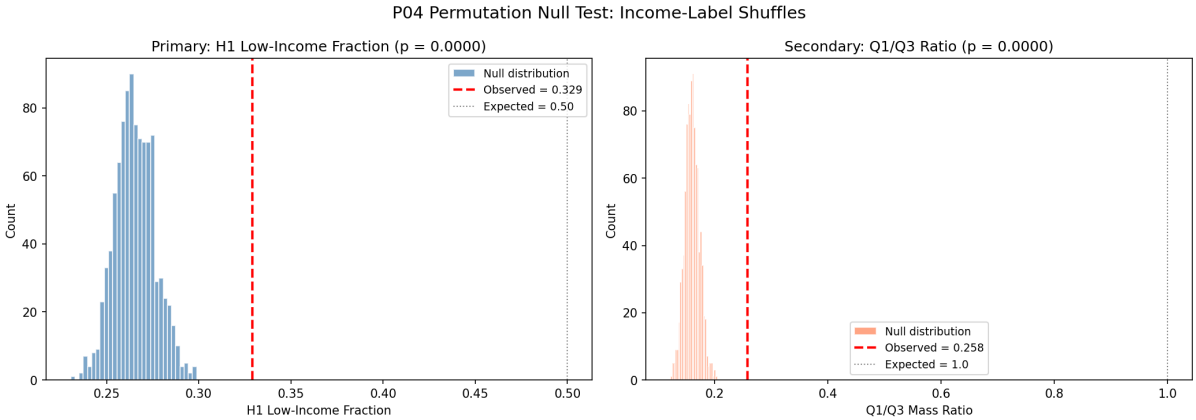

Figure 1: Permutation null distributions for H_1 low-income fraction (left) and Q1/Q3 mass ratio (right). Red dashed lines mark the observed values; grey dotted lines mark the expected values under income-exchangeability. The observed H_1 low-income fraction (0.329) falls in the upper tail of the null distribution ($p < 0.001$).

the existence of genuine topological structure (cycling patterns, closed loops) at low income that is partially masked by the dominant high-income signal.

4.4 Paper 1 Single-Parameter Baseline

Standard VR PH (single filtration parameter, ignoring income) on the same 2,000 landmarks produced the results in Table 3.

Table 3: Single-parameter VR PH on income-stratified subsets of the same 2,000 landmarks.

Subset	N	H_1 count	H_1 density
Full	2,000	2,587	1.294
Low ($\leq Q_1$)	508	481	0.947
High ($\geq Q_3$)	521	460	0.883

H_1 density is nearly identical across income subsets (0.95 vs 0.88 per point). **Single-parameter PH cannot detect the income stratification** that the bifiltration reveals. The 73.7% mass concentration at Q3/Q4 in the bifiltration is invisible to subset-level VR computation because single-parameter PH changes the complex when subsetting — it cannot decompose the joint topology.

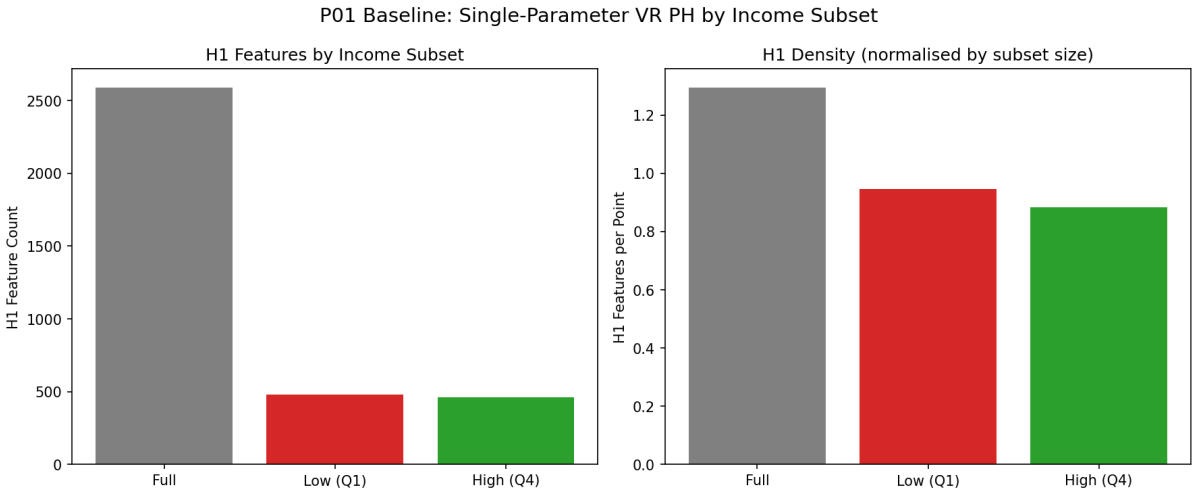

Figure 2: Paper 1 baseline comparison. Left: H_1 feature counts by income subset. Right: H_1 density (features per point), showing negligible income gradient under single-parameter PH.

4.5 GMM Regime Correspondence

The 7-component GMM from Paper 1 was applied to the 2,000 landmarks. Three regimes of note are shown in Table 4.

Table 4: Selected GMM regimes and income statistics on the landmark set.

Regime	N landmarks	Mean income	% below median
R0 (dominant)	1,720	1.05	50.4%
R5 (high-income)	16	1.55	6.2%
R6 (low-income)	99	0.47	88.9%

Regime R6 — with 88.9% of members below the median income and a mean income score of 0.47 — corresponds to the poverty trap cluster. The GMM regime decomposition is consistent with the H_1 mass gradient: trajectories in R0 and R5 (moderate-to-high income, topologically rich regions) dominate H_1 mass, while R6 trajectories occupy the topologically simpler low-income region.

4.6 Hilbert Function and Rank Invariant

Complementary invariants computed on 10×10 and 20×20 grids (detailed in Appendix B) confirm and extend the signed measure findings.

Hilbert function. The H_0 Hilbert function is income-symmetric (low-income fraction = 0.512 at both resolutions), confirming that cluster structure is balanced across

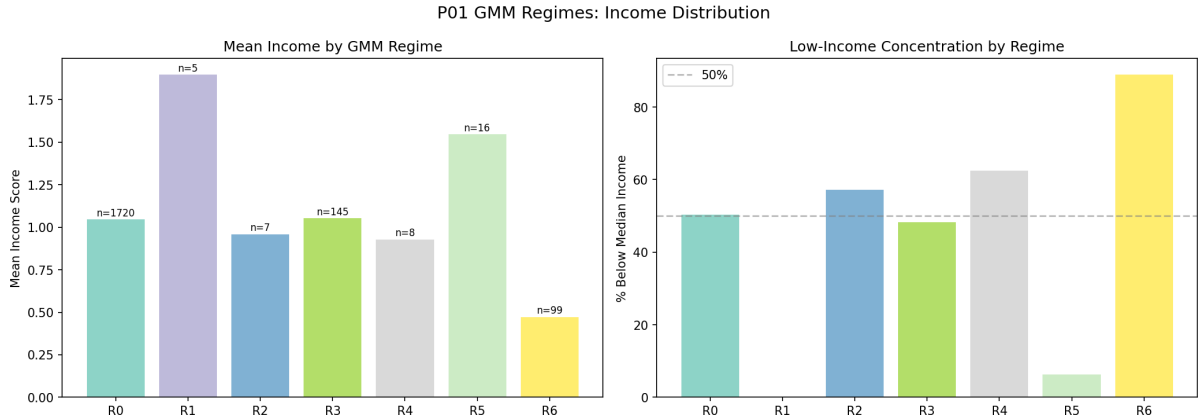

Figure 3: Paper 1 GMM regime income distributions. Left: mean income score by regime. Right: proportion of regime members below median income. Regime R6 is the poverty trap cluster (88.9% below median).

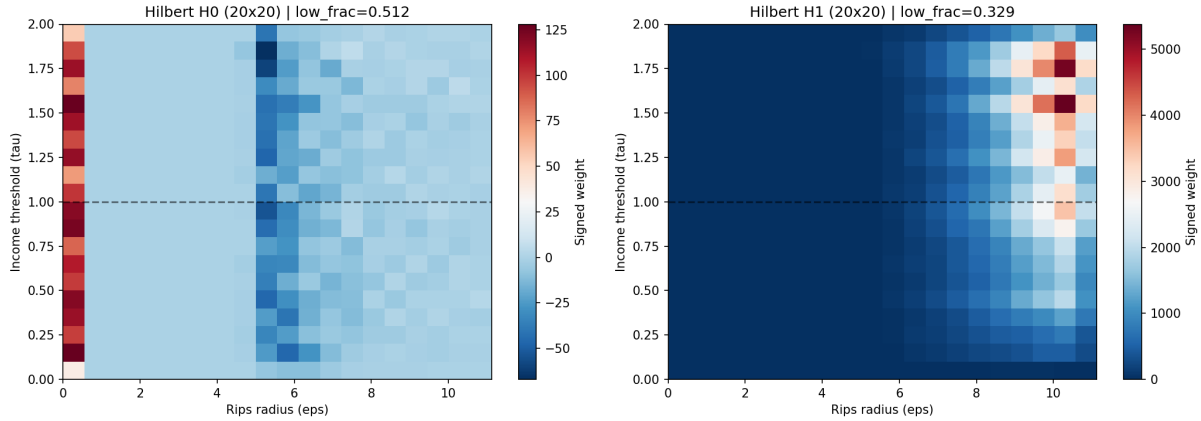

Figure 4: Hilbert function heatmaps at 20×20 resolution. Left: H_0 — income-symmetric (low-income fraction = 0.512). Right: H_1 — mass concentrated above the median income line (dashed), with near-zero weight below. The upper-right concentration confirms that loops are alive primarily at high Rips scales among high-income trajectories.

the income distribution. The H_1 Hilbert function mirrors the signed measure asymmetry with the same low-income fraction (0.329), showing a clear gradient from near-zero mass below the income median to concentrated mass in the upper-right corner of the (ε, τ) plane (Figure 4).

Rank invariant. The H_1 rank invariant on the 20×20 grid yields 218 features, of which **100% persist to infinity** — no H_1 feature born at any income threshold dies within the bifiltration range (Table 5).

This is a striking negative result. There are no income-confined topological features in the H_1 bifiltration. Every loop that enters the complex at some income threshold τ_0 remains alive as τ increases to the maximum. The income gradient in H_1 mass is

Table 5: H_1 rank invariant income persistence (20×20 grid).

Statistic	Count	Weight fraction
Total features	218	1.000
Born at low income ($\tau \leq$ median)	108	0.329
Survive to high income (of born-low)	108	1.000
Confined to low income	0	0.000
Born at Q1	53	—
Q1-born surviving to Q3	53	1.000
Persist to infinity	218	1.000

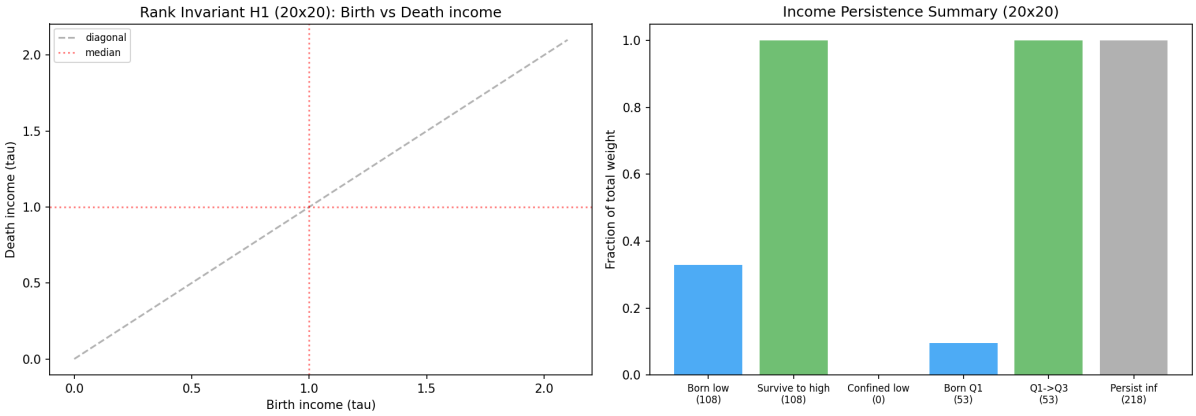

Figure 5: H_1 rank invariant analysis (20×20 grid). Left: birth vs death income scatter — empty because all features persist to infinity ($d_\tau = \infty$). Right: income persistence summary showing that 100% of features born at low income survive to high income, and 100% of all features persist indefinitely.

entirely driven by *where features are born* — predominantly at high income levels — rather than by features dying at specific income thresholds.

Interpretation for poverty traps: The absence of income-confined features means that the income gradient is monotonic rather than threshold-based. There is no critical income level at which loops appear or disappear. Instead, the topology enriches monotonically as the income threshold increases, with each new income level adding features that persist indefinitely. Low-income trajectory space is not characterised by distinct topological structures that could be "broken through"; rather, it is characterised by the *absence* of the diverse pathway structures that emerge at higher income levels.

5 Discussion

5.1 Topological Simplicity as a Poverty Trap Signature

The finding inverts a naive expectation. Rather than poverty traps manifesting as topological *features* — persistent loops or voids at low income — they manifest as topological *absence*. Low-income trajectory space is structurally simpler: fewer H_1 cycles, less multi-pathway connectivity. This makes formal topological sense: a poverty trap constrains individuals to a narrow set of employment pathways (cycling between unemployment, inactivity, and low-pay work), producing a low-dimensional, acyclic region of the embedding.

High-income trajectory space, by contrast, is topologically diverse: multiple career structures (full-time $\times$ self-employed, part-time $\times$ employed, career changes across sectors) create higher-dimensional loops in the embedding. These loops represent the career flexibility that poverty traps deny.

The rank invariant sharpens this interpretation. Because all H_1 features persist to infinity, the topology enriches monotonically with income — there is no "threshold" beyond which the structure qualitatively changes. This is more consistent with a continuous gradient of career opportunity than with a sharp poverty trap boundary. The structural constraint at low income is not a barrier that could be crossed, but rather the absence of pathway diversity that accumulates incrementally at higher income levels.

The finding is consistent with Barrett and Carter's (2013) distinction between structural and stochastic traps. Structural traps — where the constraint is on the set of reachable states — would produce exactly the topological simplicity we observe: a restricted pathway set with fewer cycles.

5.2 The Permutation Result: Real Structure Beyond Geometry

The permutation null test confirms that the income-topology association is not a geometric artefact. Under random income assignment, H_1 mass concentrates even more at high τ (null low-income fraction = 0.265 vs observed = 0.329). The real income

labels *increase* low-income H_1 mass relative to the null by approximately 24%. This means there is genuine topological structure at low income — cycling patterns among poverty-trap trajectories — that partially offsets the dominant high-income signal.

This is a two-part finding:

1. **Global asymmetry:** High-income trajectory space is inherently topologically richer (driven by the distance structure of the embedding).

2. **Income-specific moderation:** The association between income and trajectory position adds H_1 mass at low income, consistent with closed cycling patterns among trapped trajectories.

5.3 What Multi-Parameter PH Adds Beyond Single-Parameter

The baseline comparison makes the case directly: single-parameter VR PH on income-stratified subsets shows H_1 densities of 0.95 (low) and 0.88 (high) per point — effectively indistinguishable. The bifiltration's signed measure reveals a 4:1 Q3:Q1 mass ratio invisible to subset analysis. This is because single-parameter subsetting changes the simplicial complex (different points, different edges), while the bifiltration decomposes the *same* complex across income thresholds.

The Hilbert function and rank invariant add information beyond what the signed measure provides. The Hilbert function gives a "state" view — how many features are alive at each point — that visually confirms the signed measure's "differential" pattern. The rank invariant answers a question the signed measure cannot: do features born at one income level persist to another? The answer — universal persistence to infinity — constrains the class of theoretical models that could generate the observed topology.

5.4 Methodological Contribution

This paper demonstrates that multi-parameter persistent homology, via the `multipers` library's signed measure, Hilbert function, and rank invariant, is a tractable analytical

tool for panel survey data at the scale of major UK longitudinal studies. The pipeline — landmarking, bifiltration construction, signed measure computation, income-quartile decomposition, permutation null testing, Hilbert/rank diagnostics — runs in under 30 minutes on commodity hardware (i7/32 GB, no GPU). The quantile grid discretisation produces stable results across resolution settings.

5.5 Limitations

1. **Income proxy rather than monetary income:** The L/M/H income score derived from employment states is a trajectory-level proxy. Analysis with equivalised net household income (available in USoc) is planned for Paper 5.

2. **Landmark subsampling:** 2,000 of 27,280 individuals may miss rare trajectory types. Scale sensitivity analysis at 5,000 landmarks is planned before submission.

3. **Income discretisation:** The continuous τ parameter is approximated on a 20-step quantile grid. Exploratory analysis confirms stability across 10–30 steps, but effects at finer resolution cannot be ruled out.

4. **Two parameters only:** The bifiltration uses distance $\times$ income. Geography, education, or cohort as additional filtration parameters could reveal further structure but require higher-parameter methods with steeper computational cost.

5. **Single cohort pool:** The pooled 1991–2023 sample may conflate secular trends. Cohort-stratified bifiltration is a natural extension.

6. **Rank invariant resolution:** The rank invariant was computed on discrete grids (10×10 and 20×20). Features that are born and die between grid points would appear to persist to infinity. Finer-resolution grids or exact (non-gridded) computation may reveal income-confined features below the current resolution.

5.6 Future Directions

- **Paper 5:** Apply the bifiltration pipeline to SOEP (Germany), PSID (US), and CNEF harmonised data to compare income-stratified trajectory topology across welfare-state regimes.
- **Scale sensitivity:** 5,000-landmark run before submission.
- **Real income:** Comparison with equivalised net household income from USoc (Paper 5 scope).
- **Policy relevance:** If low-income topological simplicity reflects constrained career pathways, interventions that *diversify* available pathways (training, relocation support, flexible employment schemes) would increase topological complexity at low income — a measurable topological outcome.

6 Conclusion

Multi-parameter persistent homology, applied to a bifiltration indexed by Rips distance and income threshold, reveals that the topological structure of UK employment trajectories is significantly income-stratified. H_1 topological features — loops representing cyclical career structures — concentrate at mid-to-high income levels, with quartiles Q3 and Q4 carrying approximately four times the signed measure mass of Q1. Permutation null testing (999 income-label shuffles, $p < 0.001$) confirms that this is not a geometric artefact, and that real income labels increase low-income H_1 mass relative to the null by 24%.

The Hilbert function confirms this asymmetry with a complementary cumulative view, while the rank invariant shows that all H_1 features persist to infinity along the income axis — ruling out income-confined topological structures and indicating a monotonic enrichment of topology with rising income.

We interpret the topological simplicity of low-income trajectory space as a formalisation of structural poverty traps: limited career diversity constrains individuals to a

narrow set of pathways, producing less topological complexity. The rank invariant's finding of universal persistence further specifies this: the poverty trap is not a threshold to be crossed but an absence of career pathway diversity that accumulates only at higher income levels.

A Discretisation Strategy Comparison

Quantile vs fixed (uniform-spacing) grids were compared on 2,000 maxmin landmarks. Both grids produce the same qualitative finding (H_1 mass concentrates at upper income quartiles), but quantile grids show superior stability:

- H_1 low-income fraction converges to $\sim$0.33 by 10 grid steps (range $[0.33, 0.44]$) vs fixed grid (range $[0.26, 0.51]$).
- Quantile grid produces balanced H_0 stratification (51/49) vs fixed grid's high-income bias (45/55).

B Hilbert Function and Rank Invariant Analysis

B.1 Hilbert Function

The Hilbert function $\mathcal{H}_k(\varepsilon, \tau) = \dim H_k(K_{\varepsilon,\tau})$ records the number of k-dimensional features alive at each bifiltration grid point. Unlike the signed measure, which captures birth/death differentials, the Hilbert function provides a cumulative "state" view.

H_0 **Hilbert function.** Total mass = 4,074 (20×20) / 4,020 (10×10). Income distribution is symmetric (low-income fraction = 0.512 at both resolutions). The heatmap shows strong positive weights at low ε (many components before merging), turning negative at intermediate ε — the standard Rips signature of a point cloud merging into a single component. This pattern is uniform across the income axis. Quartile decomposition at 20×20: Q1 = 1,002, Q2 = 840, Q3 = 1,241, Q4 = 991.

H_1 **Hilbert function.** Total mass = 211,982 (stable across resolutions). Low-income fraction = 0.329 (identical to the signed measure). The heatmap (Figure 4) shows near-zero mass below the income median and concentrated mass in the upper-right corner (high ε, high τ), consistent with loops being alive predominantly at large Rips scales among high-income trajectories. Quartile decomposition: Q1 = 20,336 (9.6%), Q2 = 35,448 (16.7%), Q3 = 78,674 (37.1%), Q4 = 77,524 (36.6%).

Resolution stability. Hilbert function results are stable between 10×10 and 20×20 grids: total mass is identical (211,982), and low-income fraction is 0.329 at both resolutions. The 20×20 grid provides finer spatial detail in the heatmap without changing the income-stratification summary.

B.2 Rank Invariant

The rank invariant $\rho(a, b) = \text{rank}(H_k(K_a) \to H_k(K_b))$ captures which features persist from grid point a to grid point b. In `multipers`, the rank invariant signed measure returns 4D support points $(b_\varepsilon, b_\tau, d_\varepsilon, d_\tau)$ recording the birth and death coordinates in the bifiltration.

H_1 **rank invariant (**20×20**).** 218 features, total weight = 211,982. All 218 features have death coordinates at infinity ($d_\varepsilon = \infty$, $d_\tau = \infty$). See Table 5 and Figure 5 for the full income persistence breakdown.

No H_1 feature is confined to a specific income range. Every cycle that enters the bifiltration — regardless of the income threshold at which it appears — persists across the full income range. The income gradient is driven entirely by feature *birth rates* at different income levels, not by differential feature survival. This indicates monotonic topological enrichment: each successive income threshold adds new features without destroying existing ones.

10×10 **comparison.** At 10×10 resolution, 59 features were resolved (all persisting to infinity, same qualitative pattern), confirming that the universal persistence finding is not sensitive to grid resolution.

Caveat. On a discrete grid, features that are born and die between grid points

would appear to persist to infinity. Finer-resolution computation (e.g., 50×50) or exact computation on the full simplex tree could in principle reveal short-lived income-confined features below the current resolution. We consider this unlikely to change the qualitative finding, given the stability between 10×10 and 20×20, but note it as a limitation.

C Computational Notes

Code: `trajectory_tda/topology/multipers_bifiltration.py` (module), `trajectory_tda/scripts/p` (permutation runner), `trajectory_tda/scripts/p04_appendix_b.py` (Hilbert/rank runner).

Runtime: 24 minutes for 999 permutations; < 1 minute for Hilbert/rank at each resolution. All on 8-core i7/32 GB (no GPU). The `multipers` library uses sklearn fallback for distance computation (KeOps unavailable on Windows). Full results in `results/p04_multip`

Completed phases:

1. Library validation — 13/13 synthetic tests passing
2. Data pipeline — income extraction from Paper 1 sequences
3. Development-scale computation — 2,000 landmarks, quantile grid
4. Null testing — 999 permutations, $p < 0.001$
5. Paper 1 baseline comparison — H_1 density indistinguishable across income subsets
6. GMM regime overlay — R6 identified as poverty trap cluster
7. Hilbert function — H_0 symmetric, H_1 asymmetric, resolution-stable
8. Rank invariant — 100% persistence to infinity, no income-confined features

Remaining:

- 5,000-landmark scale sensitivity (before submission)

- Real income variable comparison (Paper 5 scope)

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